Data and Applications — Project Phase 3

Relational Model and Normalization

World Cube Association

Team 41

Stage A

From the Assigned EER to a Relational Schema

1.1 What we started from

We converted Team 50's final Phase 2 EER model of the WCA domain: 16 entity types (10 strong, 4 weak, and the two subclasses COMPETITOR and DELEGATE under PERSON) and 23 relationship types, including one ternary relationship (ASSIGNED_AS) and two recursive relationships (SUPERVISES, SUBREGION_OF). We applied every correction Team 50 listed in its Phase 2 submission and changed the assigned diagram only where it was demonstrably in error (Section 6).

The mapping produced **27 relations**, **39 foreign keys** and **152 columns**.

1.2 Entity types

Each strong entity became a relation of the same name with all simple attributes and the simple components of its composite attributes: CONTINENT, COUNTRY, PERSON, EVENT, FORMAT, COMPETITION, TEAM, REGION, STAFF_ROLE and RESULT. The four weak entities map to relations whose primary key is the owner's key plus the partial key: ROUND (owners COMPETITION and EVENT), REGISTRATION (owner COMPETITION), SCRAMBLE (owner ROUND) and ATTEMPT (owner RESULT).

1.3 Specialization

PERSON is specialized into COMPETITOR and DELEGATE. The specialization is overlapping and partial, so we used the multiple-relations option (textbook Option 8A): one relation for the superclass and one per subclass, each subclass keyed by Person_No as a foreign key to PERSON. COMPETITOR is predicate-defined: a COMPETITOR row exists exactly when PERSON.WCA_ID is not null. A single combined relation was rejected because it would force nulls for subclass-specific attributes, and a total-specialization mapping does not fit a partial specialization.

1.4 Relationships

Binary 1:N and 1:1 relationships became foreign keys on the "many" (or total) side; M:N relationships and the ternary relationship became their own relations.

Relationship(s)	Mapping
IN_CONTINENT , REPRESENTS (+ Since_Date), HOSTED_IN , USES_FORMAT , SERVES_IN , SUBREGION_OF , MAIN_EVENT	Foreign key on the many / total side
HAS_ROUND + FOR_EVENT	The ROUND key columns (Competition_ID , Event_ID)
RECEIVES , SUBMITS	REGISTRATION.Competition_ID , REGISTRATION.Person_No
ACHIEVES , IN_ROUND , REPRESENTED_AT	RESULT.Person_No , RESULT.(Competition_ID, Event_ID, Round_Number) , RESULT.Country_ID
CONSISTS_OF , SCRAMBLED_FOR	ATTEMPT.Result_ID , SCRAMBLE roundkey columns
ALLOWS , SELECTS , DELEGATES , ORGANIZES , MEMBER_OF (+ Role , Start /End_Date)	Separate M:N relations
ASSIGNED_AS (ternary: PERSON , ROUND , STAFF_ROLE)	ASSIGNED_AS relation + ASSIGNED_AS_SLOTS
SUPERVISES , SUBREGION_OF (recursive)	DELEGATE.Supervisor_Person_No , REGION.Parent_Region_ID

1.5 Composite attributes

All nine composite attributes were split into simple columns during mapping, so no composite value survives into any relation.

Composite	Columns
PERSON.Name	Romanized_Name , Local_Name
COMPETITION.Venue	Venue_Name , Street_Address , Venue_Details , City
COMPETITION.Coordinates	Latitude , Longitude
COMPETITION.Date_Range	Start_Date , End_Date
COMPETITION.Registration_Window	Opens_At , Closes_At
COMPETITION.Entry_Fee	Amount , Currency_Code
ROUND.Time_Limit	Per_Attempt (+ CUMULATIVE_SCOPE for cumulative limits)
ROUND.Cutoff	Cutoff_Attempts , Cutoff_Value
ROUND.Advancement	Adv_Type , Adv_Level

1.6 Multi-valued attributes

Each of the six multi-valued attributes was mapped under the option matching its bound.

Attribute	Option	Result
ALLOWS.Usage	4(a)	Fixed columns (max 2, domain {full_round, cutoff_phase}): Usage_1 (NOT NULL), Usage_2 (nullable) — assumption A7
PERSON.Previous_Identities	4(b)	PREVIOUS_IDENTITIES
COMPETITION.Championship_Types	4(b)	CHAMPIONSHIP_TYPES
ROUND.Cumulative_Scope	4(b)	CUMULATIVE_SCOPE (each value = two columns)
SCRAMBLE.Sequences	4(b)	SCRAMBLE_SEQUENCES
ASSIGNED_AS.Slots	4(b)	ASSIGNED_AS_SLOTS

1.7 Derived attributes and controlled redundancy

Twelve derived attribute groups are not stored (computed on read): PERSON.Competition_Count, Medals (Gold/Silver/Bronze) and Record_Count (WR/CR/NR); COMPETITOR.First_Year; COMPETITION. Duration_Days, Status and Spots_Left; REGION.Level; ROUND.Round_Type, Scramble_Set_Count and Competitor_Count; ATTEMPT.MBLD_Decoded (Solved/Attempted/Seconds).

Five attributes are stored in RESULT as trigger-maintained controlled redundancy, exactly as the WCA export keeps them for ranking performance: Best, Average, Position, Single_Record, Average_Record (T54 §4.5). These are not used to derive normalization FDs.

1.8 The Stage A relations

Every column not marked “?” is NOT NULL. “Candidate keys” lists keys beyond the primary key.

#	Relation	Cols	Primary key	Other candidate keys / uniqueness	FKs
1	CONTINENT	3	Continent_ID	{Name} (UNIQUE); Record_Code not a key	—
2	COUNTRY	4	Country_ID	{ISO2} (UNIQUE NOT NULL)	F01
3	PERSON	9	Person_No	WCA_ID unique among non-nulls (not a key)	F02
4	PREVIOUS_IDENTITIES	4	all four columns	—	F03, F04
5	COMPETITOR	1	Person_No	—	F05
6	DELEGATE	7	Person_No	{WCA_Email} (UNIQUE stated; NOT NULL by A3)	F06–F08
7	EVENT	5	Event_ID	{Event_Name}	—
8	FORMAT	5	Format_ID	{Format_Name}	—

9	ALLOWS	4	Event_ID, Format_ID	—	F09, F10
10	COMPETITION	23	Competition_ID	—	F11, F12
11	CHAMPIONSHIP_TYPES	2	both columns	—	F13
12	TEAM	4	Team_ID	—	—
13	MEMBER_OF	5	Person_No, Team_ID	—	F14, F15
14	REGION	3	Region_ID	—	F16
15	STAFF_ROLE	3	Role_Code	—	—
16	ROUND	12	Competition_ID, Event_ID, Round_Number	—	F17–F20
17	CUMULATIVE_SCOPE	5	all five columns	—	F21
18	SCRAMBLE	6	all six columns	—	F22
19	SCRAMBLE_SEQUENCES	7	all seven columns	—	F23
20	REGISTRATION	7	Competition_ID, Registrant_No	{Competition_ID, Person_No}	F24, F25
21	SELECTS	3	all three columns	—	F26, F27
22	RESULT	11	Result_ID	{Competition_ID, Event_ID, Round_Number, Person_No}	F28–F30
23	ATTEMPT	3	Result_ID, Attempt_No	—	F31
24	DELEGATES	2	both columns	—	F32, F33
25	ORGANIZES	2	both columns	—	F34, F35
26	ASSIGNED_AS	5	all five columns	—	F36–F38
27	ASSIGNED_AS_SLOTS	7	Person_No, Competition_ID, Event_ID, Round_Number, Role_Code, Group_ID	—	F39

Column total: 3 + 4 + 9 + 4 + 1 + 7 + 5 + 5 + 4 + 23 + 2 + 4 + 5 + 3 + 3 + 12 + 5 + 6 + 7 + 7 + 3 + 11 + 3 + 2 + 2 + 5 + 7 = 152.

Assumptions surfaced here: **A4** (Station_No depends on the whole six-column slot key), **A5** (Per_Attempt carries either the per-attempt or the cumulative limit), **A7** (fixed Usage columns), **A8** (cumulative-scope round is same-competition and trigger-checked). The full register is in the Appendix.

Stage B

First Normal Form

No conversion was needed. All 27 relations hold atomic values: the composite attributes of PERSON, COMPETITION and ROUND were split into simple columns in Stage A, the multi-valued attributes were

mapped using the permitted 4(a)/4(b) approaches, and all stored columns contain atomic values. The one fixed-column case, ALLOWS.Usage_1/Usage_2, is the Stage A item 4(a) form the specification explicitly accepts as 1NF; the five 4(b) relations (PREVIOUS_IDENTITIES, CHAMPIONSHIP_TYPES, CUMULATIVE_SCOPE, SCRAMBLE_SEQUENCES, ASSIGNED_AS_SLOTS) each store one atomic value per row. The structured-looking source attributes WCA_ID, Location, Championship_Type, Sequence and Value are single opaque values. The model is already in 1NF and carries forward unchanged (27 relations, 152 columns). **No Stage B snapshot is required** (P3 §1.2).

Stage C

Second Normal Form

No conversion was needed. All 27 relations are in 2NF: the 11 relations with single-column candidate keys cannot have partial dependencies, the 9 all-key relations have no non-prime attributes,

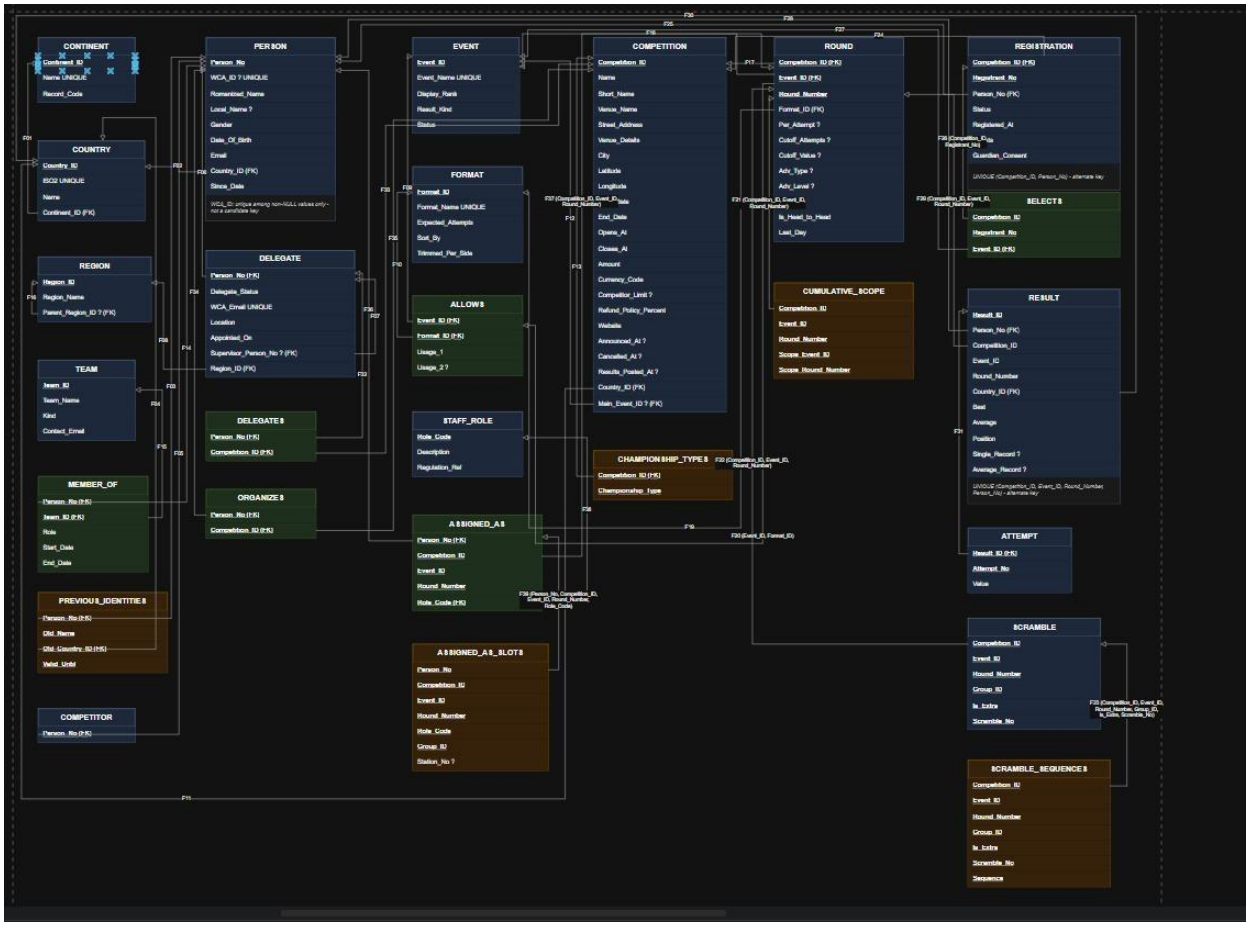


Figure 1. Stage A relational schema.
https://drive.google.com/file/d/1ZKZ9kOYsBKZ2a6DW4jhirIwOEYFXnUS0/view?usp=sharing

and the remaining 7 relations were tested against every maximal proper subset of their candidate keys without finding a partial dependency. The detailed tests supporting this statement appear in Section 5.2.

Same Diagram For B,C .

Reason:
Stage B: First Normal Form

No conversion was needed. All 27 relations are already in 1NF because every attribute contains atomic values. Composite attributes were decomposed into simple columns during Stage A, and multivalued attributes were mapped using the permitted 4(a) and 4(b) approaches. Therefore, the Stage A relational model already satisfies 1NF, with 27 relations and 152 columns. As no conversion was required, a separate Stage B snapshot is not submitted.

Stage C: Second Normal Form

No conversion was needed. All 27 relations are already in 2NF. The 11 relations with single-attribute candidate keys cannot have partial dependencies, while the 9 relations whose attributes are all part of candidate keys have no non-prime attributes. The remaining 7 relations with composite candidate keys and non-prime attributes were checked for partial dependencies, and none were found. Therefore, the Stage B relational model already satisfies 2NF, with 27 relations and 152 columns. As no conversion was required, a separate Stage C snapshot is not submitted.

As stage D starts from Stage C, not independently from Stage A, Since B and C required no changes, the Stage D model is derived from the same 27-relation, 152-column model, with only the required 3NF change: removing DELEGATE.Location.

Stage D

Third Normal Form

One change. In DELEGATE, Region_ID → Location holds because Location is the name of the Delegate's region (Team 50 §5.1–§5.2, interpretation A1). Region_ID is not a superkey of DELEGATE (the candidate keys are {Person_No} and {WCA_Email}), and Location is non-prime, so Person_No →Region_ID→Location is a transitive dependency and violates 3NF.

Resolution. We removed Location from DELEGATE and created no new relation, because REGION already stores the determinant: REGION.Region_Name holds the value and DELEGATE.Region_ID → REGION.Region_ID (F08) links to it.

- **Lossless join:** Region_ID is REGION's key and a NOT NULL foreign key in DELEGATE, so DELEGATE⍰⍰REGION on Region_ID reconstructs the former Location as Region_Name with no row loss and no spurious rows.
- **Dependency preservation:** Region_ID → Region_Name remains in REGION and every other DELEGATE dependency stays in DELEGATE.

The other 26 relations were checked and need no change. **Result:** 27 relations, 39 foreign keys, 151 columns (DELEGATE drops from 7 to 6 columns). This is a normalization decision, not an EER change — the assigned diagram keeps Location.

5 Functional Dependencies and Candidate Keys (all 27 relations)

Dependencies are stated in the form A, B → C and come from the meaning of the assigned data (P3 §3.4). "rest" below denotes the relation's remaining columns.

5.1 Accepted dependencies and candidate keys

Relation	Candidate keys	Non-trivial FDs
CONTINENT	{Continent_ID}, {Name}	each key ⍰ rest
COUNTRY	{Country_ID}, {ISO2}	each key ⍰ rest
PERSON	{Person_No}	Person_No ⍰ rest
PREVIOUS_IDENTITIES	all four columns	none non-trivial (all-key)
COMPETITOR	{Person_No}	none non-trivial
DELEGATE	{Person_No}, {WCA_Email}	each key ⍰ rest; Region_ID ⍰ Location (A1, Stage A–C only; removed at Stage D)
EVENT	{Event_ID}, {Event_Name}	each key ⍰ rest
FORMAT	{Format_ID}, {Format_Name}	each key ⍰ rest
ALLOWS	{Event_ID, Format_ID}	key ⍰ Usage_1 , Usage_2
COMPETITION	{Competition_ID}	key ⍰ rest
CHAMPIONSHIP_TYPES	all columns	all-key
TEAM	{Team_ID}	key ⍰ rest
MEMBER_OF	{Person_No, Team_ID}	key ⍰ Role , Start_Date , End_Date
REGION	{Region_ID}	key ⍰ rest
STAFF_ROLE	{Role_Code}	key ⍰ rest
Relation	**Candidate keys**	**Non-trivial FDs**
ROUND	{Competition_ID, Event_ID, Round_Number}	key ⍰ all other columns
CUMULATIVE_SCOPE	all five columns	all-key
SCRAMBLE	all six columns	all-key
SCRAMBLE_SEQUENCES	all seven columns	all-key

REGISTRATION	{Competition_ID, Registrant_No}, {Competition_ID, Person_No}	each key → rest
SELECTS	all three columns	all-key
RESULT	{Result_ID}, {Competition_ID, Event_ID, Round_Number, Person_No}	each key → rest
ATTEMPT	{Result_ID, Attempt_No}	key → Value
DELEGATES, ORGANIZES, ASSIGNED_AS	all columns	all-key
ASSIGNED_AS_SLOTS	{Person_No, Competition_ID, Event_ID, Round_Number, Role_Code, Group_ID}	key → Station_No (A4)

Dependencies labelled from assumptions: **A1** (Region_ID → Location), **A4** (slot key → Station_No), and **A7** (fixed-column representation of Usage). The remaining key-to-rest dependencies follow from the accepted candidate keys and the meanings of the assigned attributes.

5.2 2NF evidence — the seven composite-key relations with non-prime attributes

For a key of n columns it suffices to test the n maximal $(n-1)$-column subsets: if none determines a non-prime attribute, augmentation guarantees no smaller subset does. A subset is shown not to determine an attribute by a scenario the assigned documents permit in which two rows agree on the subset but differ on the attribute.

ALLOWS key {Event_ID, Format_ID}; non-prime Usage_1, Usage_2. Usage sets from T54 Tables 1 and 9: (333fm, Best of 1) = {full_round, cutoff_phase}, (333fm, Mean of 3) = {full_round}, (666, Mean of 3) = {full_round}, (666, Best of 1) = {cutoff_phase}.

Subset	Counterexample	Result
{Event_ID}	333fm with Best of 1 vs Mean of 3 differ in Usage_2; 666 with Mean of 3 vs Best of 1 differ in Usage_1	determines neither
{Format_ID}	Best of 1 gives {full_round, cutoff_phase} with 333fm but {cutoff_phase} with 666	determines neither

MEMBER_OF key {Person_No, Team_ID}; non-prime Role, Start_Date, End_Date.

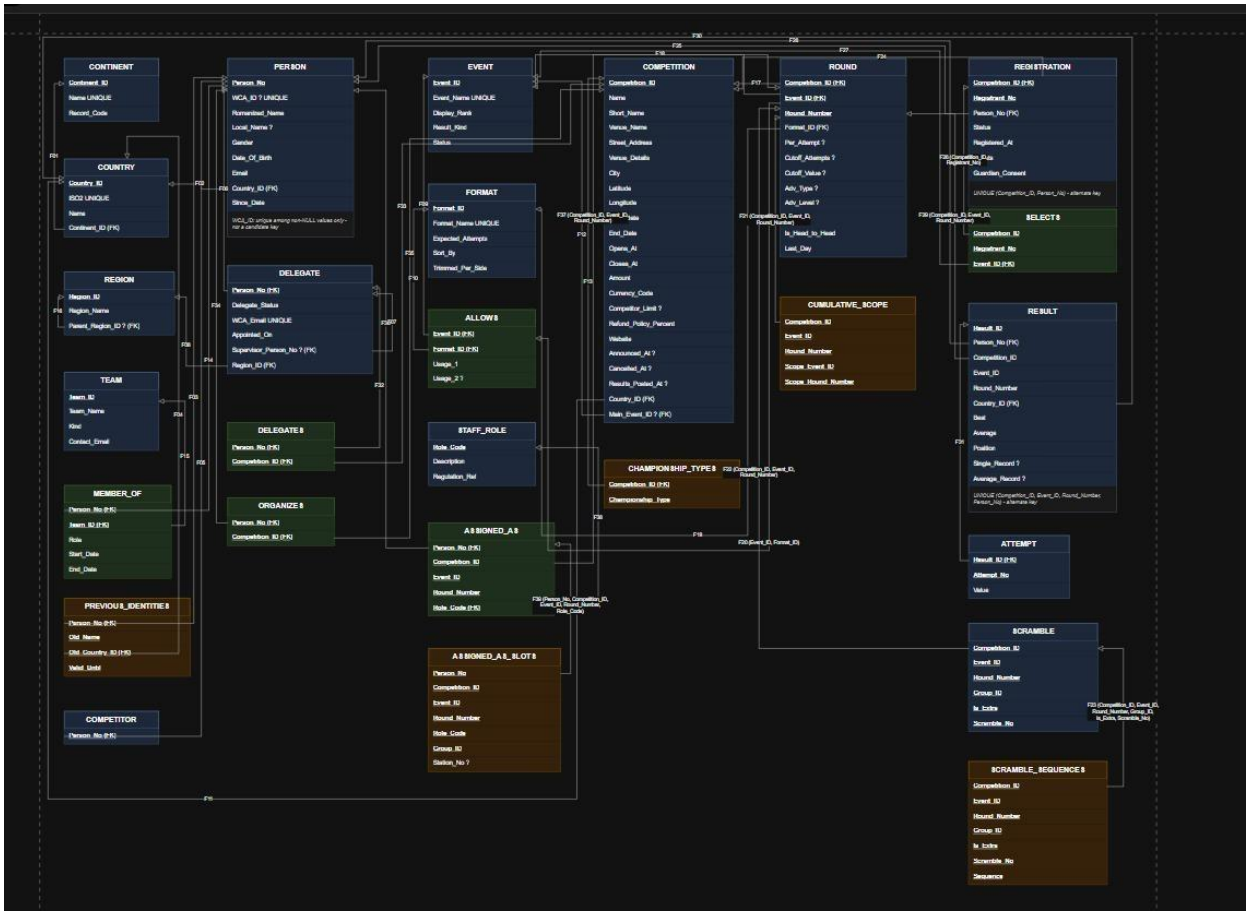


Figure 2. Stage D relational schema (DELEGATE.Location removed).
https://drive.google.com/file/d/1ZKZ9kOYsBKZ2a6DW4jhirIwOEYFXnUS0/view?usp=sharing

Subset	Counterexample	Result
{Person_No}	A person leads one team and is an ordinary member of another, with different dates (M:N, T54 rel. 23)	determines none
{Team_ID}	One team has many members with different roles / dates	determines none

ROUND key {Competition_ID, Event_ID, Round_Number}; non-prime Format_ID , Per_Attempt , Cutoff_Attempts , Cutoff_Value , Adv_Type , Adv_Level , Is_Dual , Is_Head_to_Head , Last_Day . T54 §1.4: each round announces its own format, limit, cutoff and advancement; cross-round equalities apply only to Dual Rounds (C18).

Subset (agree on)	Differ in	Attributes shown not determined
{Competition_ID, Event_ID}	Round_Number	Adv_Type /Adv_Level (null for final, Table 15), Is_Head_to_Head (C17), Is_Dual (C18), Last_Day , Format_ID , Per_Attempt , Cutoff_* (§1.4)

{Competition_ID, Round_Number}	Event_ID	Format_ID , Cutoff_Attempts (333 Avg-of-5/Best-of-2 vs 666 Mean-of-3/Best-of-1, Table 1), Cutoff_Value (cs vs moves), Is_Head_to_Head (C17), rest event-specific
{Event_ID, Round_Number}	Competition_ID Last_Day	(Table 10), Adv_Type /Adv_Level (sizedependent, C8), rest per-competition (§1.4)

REGISTRATION eys {Competition_ID, Registrant_No} and {Competition_ID, Person_No}; no prime Status , Registered_At , Guests , Guardian_Consent .

Key	Subset	Counterexample	Result
{Comp, Registrant_No}	{Competition_ID}	many registrations per competition (RECEIVES 1:N)	none
	{Registrant_No}	Registrant_No restarts per competition (T54 Table 16)	none
{Comp, Person_No}	{Competition_ID}	as above	none
	{Person_No}	a person registers for many competitions; Guardian_Consent depends on age at each Start_Date	none

RESULT keys {Result_ID} and {Competition_ID, Event_ID, Round_Number, Person_No}; non-prime Best, Average, Position, Single_Record, Average_Record, Country_ID. {Result_ID} is single-column; only the four-column key is tested.

Subset (agree on)	Differ in	Result
{Comp, Event, Round}	Person_No	different people different results / positions / countries
{Comp, Event, Person}	Round_Number	one result per person per round
{Comp, Round, Person}	Event_ID	different events different results
{Event, Round, Person}	Competition_ID	different competitions different results

Country_ID note (decision U11): T54 rel. 14 and §4.5 record the country on the result day, and C16 only limits how often it may change; no source fixes it within a competition or event, so Competition_ID, Person_No→Country_ID is **rejected**. No partial dependency exists. This keeps RESULT's Country_ID and Position; the two decisions must remain consistent, since accepting that FD would make the second row a 2NF violation.

ATTEMPT key {Result_ID, Attempt_No}; non-prime Value.

Subset	Counterexample	Result

{Result_ID}	a result has 1–5 attempts with different values (CONSISTS_OF (1,5))	does not determine Value
{Attempt_No}	attempt 1 of two results differs	does not determine Value

ASSIGNED_AS_SLOTS key {Person_No, Competition_ID, Event_ID, Round_Number, Role_Code, Group_ID};

non-prime Station_No . All six five-column subsets differ on Station_No :

Dropped column	Why Station_No can differ
Group_ID	one person judges groups A and C at different stations (§4.6.2)
Person_No	two judges in one group use different stations
Competition_ID	station layouts are competition-specific
Event_ID	333-r1 group A vs 444-r1 group A differ
Round_Number	333-r1 vs 333-r2 group A differ
Role_Code	same person judges and scrambles a group at different stations (§4.6.2)

Station_No depends on the whole six-column key (A4). No partial dependency.

5.3 Dependencies checked and rejected

Relation	Dependency	Reason rejected
CONTINENT	Record_Code □ Continent_ID, Name	uniqueness never stated (T54 Table 5 kind S; C1) — A2 withdrawn, U4
PERSON	WCA_ID □ rest (relationwide)	WCA_ID is nullable
FORMAT	Expected_Attempts, Sort_By □ rest	not tabulated per format; formats can be added
RESULT	Competition_ID, Person_No □ Country_ID	country recorded per result day (U11)
RESULT	RK, Best, Average □ Position	Dual/Head-to-Head ranking rules differ (U12)
DELEGATE	Region_ID □ Supervisor_Person_No	no source
ALLOWS	Usage_2 □ Usage_1	artefact of null padding
ALLOWS	Event_ID □ Usage ; Format_ID □ Usage	Usage varies by the other attribute
COMPETITION	City □ Country_ID ; Lat,Long □ venue; Country_ID □ Currency_Code	no assigned rule
MEMBER_OF	Team_ID, Role □ Person_No	many people share a role
PERSON	Email □ Person_No	no uniqueness rule for Email
SCRAMBLE_SEQUENCES	Sequence □ scramble key	no uniqueness rule

6 Change to the Assigned EER

MAIN_EVENT. In Team 50's final diagram the MAIN_EVENT diamond is joined to the FOR_EVENT identifying diamond rather than to EVENT. We connected it to EVENT, as defined by Team 54 relationship 6 and Team 50 Appendix A (R6: COMPETITION (0,1) – EVENT (0,N)), giving COMPETITION. Main_Event_ID → EVENT.Event_ID (foreign key F12). This is the only change we made to the assigned EER.

7 Generative-AI Usage Statement

We used Gen AI during the preparation of this report. It helped us organize the report sections, improve formatting, prepare and debug our LaTeX source, and cross-check information on the official WCA website with our Phase 2 document.

The database design work was carried out by our team. This included mapping the ER diagram, identifying and verifying keys, working through the 1NF, 2NF and 3NF normalization steps, and making the U1–U12 design decisions. These decisions and analyses were based on our understanding of the project requirements.

Appendix A — Assumptions, Decisions and Constraints

Assumptions (A1–A9), A2 withdrawn

A1 Location = linked Region_Name; **A3** WCA_Email NOT NULL (UNIQUE is stated); **A4** one station per person/round/role/group; **A5** Per_Attempt carries per-attempt or cumulative limit, not both; **A6** external WCA evidence used only to clarify, never to add schema; **A7** fixed Usage columns; **A8** cumulative-scope round is same-competition, trigger-checked; **A9** no minimum-one-slot rule. **A2**

(Record_Code unique/candidate key) is withdrawn: T54 Table 5 classifies Record_Code as a label (kind S) and C1 does not establish uniqueness.

Key decisions (U1–U12)

U1 WCA_ID nullable-unique, not a key. **U2** accept Region_ID → Location, apply at Stage D.

U4 Record_Code not a candidate key. **U5** FORMAT keys {Format_ID}, {Format_Name}. **U6** PREVIOUS_IDENTITIES four-column key. **U7** six-column ASSIGNED_AS_SLOTS key, Station_No non-key.

U8 WCA_Email UNIQUE + NOT NULL (A3) ⍰ candidate key. **U9** no added uniqueness for Display_Rank/Regulation_Ref/Team_Name/Contact_Email/Region_Name. **U10** no Cumulative_Limit column. **U11** reject Comp,Person→Country_ID; keep RESULT.Country_ID. **U12** reject RK,Best,Average →Position; keep RESULT.Position.

Domain constraints

Domain constraints carry over from T54 (verify wording against T54 Tables 5–18 before presenting as implemented): continent-record codes, ISO2 two-letter, WCA_ID pattern, Delegate status/email domain, event/format/round/registration/result domains, coordinate and date-ordering checks.

Stage snapshot summary

Stage	Relations	FKs	Columns	Conversion	Snapshot
A	27	39	152	ER ? relational mapping	required
B	27	39	152	none (1NF)	not required
C	27	39	152	none (2NF)	not required
D	27	39	151	remove DELEGATE.Location	required